

3(a)	any three from: decreases / reduces, transpiration ; A lower transpiration rate <i>accept in the correct context once only</i> <i>counts towards max two</i> I evaporation <i>max two for cuticle</i> thick (waxy) cuticle ; increased waterproof layer ; increased (diffusion) distance for <u>water vapour</u> / less <u>water vapour</u> lost ; <i>in context of cuticular transpiration</i> <i>max two for stoma</i> sunken stoma ; A stoma in depression / AW moist / humid, air collects in area near to external environment ; AW decreases water potential gradient ; <i>in context of between sub-stomatal air space or area above stoma and external environment</i> <i>alternative acceptable statement for the transpiration mark</i> reduces <u>diffusion</u> of <u>water vapour</u> (out) via the stoma ;	3
3(b)(i)	any three from: (many) mitotic cell cycles / mitoses ; AW <i>ref. to division of (meri)stem cell also produces more (meri)stem cells ;</i> formation of rootlets by mitosis is, growth / increase in number of cells ; <i>idea of producing large number of cells</i> R cell growth differentiation / specialisation ; <i>in context of the new cells formed by stem cell division</i>	3

Question	Answer	Marks
3(b)(ii)	any three from: active transport occurring ; (pumped / moved, out) against the(ir) concentration gradient ; (move out through), carrier / pump, protein ; R channel protein using, ATP / energy ; A have ATPase activity AVP ; e.g. membrane impermeable to entry of anions suggestion that move out by exocytosis <i>allow ATP mp if also stated;</i>	3
3(b)(iii)	any two from: increased surface area ; (because, many / more) root hairs / root hair cells ; increase in, carrier proteins / channel proteins (for ion uptake) ; A transport proteins / aquaporins (for water) R channel proteins AVP ; increase in water uptake osmotically to follow ion uptake / AW ;	2

Question	Answer	Marks
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